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9) Preparation of alkyl cyanide

Grignard reagent reacts with cyanogen chloride to from alkyl cyanide

Example

$CH_3MgI + CNCl \longrightarrow CH_3CN + Mg(Cl)(I)$

Cyanogen chloride — Methyl cyanide

10) Preparation of Alkanes

Compounds like water, alcohols and amines which contain active hydrogen atom react with Grignard reagents to form alkanes.

Example

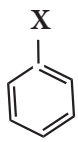
$CH_3MgI + HO-H \longrightarrow CH_4 + MgI(OH)$

$CH_3MgI + C_2H_5OH \xrightarrow{\Delta} CH_4 + MgI(OC_2H_5)$

Ethyl alcohol — methane

14.5 Haloarenes

Haloarenes are the compounds in which the halogen is directly attached to the benzene ring.

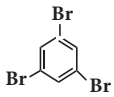
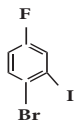
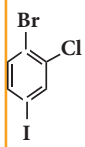


14.5.1. Nomenclature of haloarenes

In the IUPAC nomenclature, the halo arenes are named by adding prefix halo before the name of the aromatic hydrocarbon. For naming disubstituted arenes, the relative position of the substituent 1,2; 1,3 and 1,4 are indicated by the prefixes ortho, meta and para, respectively.

For poly haloarenes the numbering should be done in such a way that the lowest possible number should be given to the substituents and the name of the halogens are arranged in alphabetic order.

Nomenclature can be well understood from the following examples.

Compound	IUPAC NAME
1	 1, 2 - Dichlorobenzene
2	 1, 3- Dibromobenzene
3	 1,3,5-tri bromobenzene
4	 1- Bromo - 4- fluoro -2- iodobenzene
5	 1- Bromo -2- chloro-4- iodobenzene

14.5.2 Nature of C- X bond in haloarenes

In halo arenes the carbon atom is sp^2 hybridised. The sp^2 hybridised orbitals are shorter and holds the electron pair of bond more tightly.

Halogen atom contains P-orbital with lone pair of electrons which interacts with π-orbitals of benzene ring to form extended conjugated system of π- orbitals. The delocalisation of these electrons give double bond character to C – X bond. The resonance structure of halobenzene is given as

